

Many of the advances and ideas put forward by the ancient Greeks were lost in the centuries that followed. In many cases, it was not until relatively modern times that they were considered again.

ROMAN SCIENCE

Greek scientific knowledge was used by the Romans, although much was later lost and they tended not to be great theoretical thinkers.

ARAB SCIENTISTS

The ideas of the Greek natural philosophers passed to Arab scholars. In particular, the **Arabs translated the works of Aristotle into Arabic**. Great Arabic philosophers, such as Avicenna and Averroes, extended classical Greek thinking and added their own ideas and observations **174–77 >>**

THE CATHOLIC CHURCH

Aristotle's flawed theories were **accepted as fact** by the Catholic Church in Europe, which dramatically set back scientific progress.

MEDIEVAL SCHOLARS

Geographical knowledge was revived with maps of the world (*Mappa Mundi*). This 11th-century example is from Winchester.

SCIENTIFIC REVOLUTION

In the 17th and 18th centuries, many **new thinkers** revived the spirit of scientific approach **266–67 >>**

NEW NATURAL PHILOSOPHY

Today, ancient Greek ways of thinking about matter and energy are still relevant; scientists of physics and chemistry ask much the same questions as Thales, Plato, and Aristotle.



MAPPA MUNDI

The weight of the world

Legend has it that the Greek god Atlas was made by Zeus to hold up the universe as a punishment. This was the sort of idea that the Greek thinkers began to question, with their rational, scientific investigations and theories.

Aristotle (384–322 BCE), a pupil of Plato. Aristotle reversed his teacher's idea about the role of mathematics. He maintained that truth was to be found only in reality—an idea known as empiricism. The theories of both remain important in scientific thinking today.

Using logic and observation, Plato and Aristotle deduced many previously unknown scientific facts. For example, they proved that Earth is round, not flat. Aristotle was perhaps the most prolific of the Greek natural philosophers. He wrote texts on a range of subjects, including botany, zoology, astronomy, anatomy, and physics. Aristotle's theories were plausible and well thought out, but were based on common sense, rather

than rigorous logic and careful observation. Unfortunately, many were later discovered to be seriously flawed.

Later developments

During the final phase of the ancient Greek period in the 3rd and 2nd centuries BCE, great thinkers began to come full circle, applying the tools of Greek natural philosophy—mathematics and logic—to practical and technical challenges. For example, Archimedes (c. 287–c. 212 BCE) was a brilliant theoretical mathematician, but also a great engineer and inventor. Further developments were made at institutions such as the Museum and Library at Alexandria. Competition between centers of learning led to further innovations and the development of science.

MATHEMATIKA The ancient Greek word for “mathematics” is derived from a more general Greek word, *mathema*, which means “learning” or “study.”

INVENTION

HIPPOCRATIC OATH

The ancient Greeks' scientific way of thinking extended to the study of medicine. More than anyone, the physician Hippocrates (c. 460–c. 370 BCE), promoted a rational, scientific approach to medicine, giving it a firm footing as a professional endeavor. He placed importance on the careful observation of symptoms, and rejected traditional temple medicine theories about illnesses and their cures.

Even today, the ethical “contract” between doctors and patients is known as the Hippocratic Oath. We do not know whether Hippocrates was the author of the original, with promises to work “for the good of my patients according to my ability and my judgment and never do harm to anyone.” It is preserved on papyri of later periods, such as this 3rd-century example discovered in Egypt.

